

Symmetry: Even & Odd Functions

Name: _____

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Even function: symmetric across the _____; algebraically $f(-x) = \underline{\hspace{2cm}}$.

Odd function: symmetric about the _____; algebraically $f(-x) = \underline{\hspace{2cm}}$.

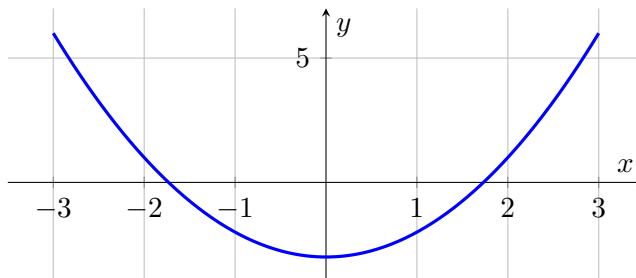
We do together. Is $f(x) = x^2 - 3$ even, odd, or neither?

Step 1. Replace x with $-x$: $f(-x) = (-x)^2 - 3 = \{\{x^2 - 3 - 2\text{cm}\}\}$.

Step 2. Compare to $f(x)$: they are _____, so $f(-x) = f(x)$.

Step 3. Therefore f is _____ (symmetric across the y -axis).

See it (even = y -axis mirror):



$y = x^2 - 3$ — the y -axis is a mirror line (even).

You Try. Solve each. Show every step, then name the answer.

a) Is $f(x) = x^3$ even, odd, or neither?

b) Is $f(x) = x^2 + x$ even, odd, or neither?

Summary (fill in): Compute $f(-x)$: equals $f(x) \Rightarrow$ _____; equals $-f(x) \Rightarrow$ _____; otherwise _____.